Subject: Mathematics **WEEKLY CHAPTER TEST – 02 (PROBABILITY)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 14: Probability** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 1 VSAQ question (2 marks). Section C has 3 SA questions (3 marks each).
3. Section D has 1 Applied Case Study (4 marks). Show all working clearly.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. If $P(E) = 0.05$, what is the probability of 'not E' (i.e. $P(\bar{E})$)? **[1]**

Options: (a) 0.95 (b) 0.05 (c) 1.05 (d) 0.095

2. A single die is thrown once. The probability of getting a prime number is: **[1]**

Options: (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{1}{6}$

3. From a standard deck of 52 cards, the probability of drawing a queen of clubs or a king of hearts is: **[1]**

Options: (a) $\frac{1}{13}$ (b) $\frac{1}{26}$ (c) $\frac{1}{52}$ (d) $\frac{2}{13}$

4. Cards numbered from 1 to 20 are mixed thoroughly and a card is drawn at random. The probability that the drawn card bears a number which is a multiple of 3 is: **[1]**

Options: (a) $\frac{3}{10}$ (b) $\frac{1}{3}$ (c) $\frac{3}{20}$ (d) $\frac{7}{20}$

5. **Assertion (A):** If the probability of winning a game is 0.4, then the probability of losing it is 0.6.
Reason (R): For any event E, $P(E) + P(\bar{E}) = 1$, where $\bar{E}$ represents the negation/complement of event E. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

6. Two coins are tossed simultaneously. Write all possible outcomes and find the probability of getting exactly one tail. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

7. A bag contains 5 red balls, 8 white balls, 4 green balls, and 7 black balls. If one ball is drawn at random, find the probability that it is: (i) black (ii) red or white (iii) not green. **[3]**

8. Cards numbered 1 to 30 are put in a bag. A card is drawn at random from this bag. Find the probability that the drawn card has: (i) a prime number greater than 10 (ii) a number divisible by both 2 and 3 (iii) a number not divisible by 5. **[3]**

9. Five cards—the ten, jack, queen, king, and ace of diamonds—are well-shuffled with their face downwards. One card is then picked up at random. (i) What is the probability that the card is the queen? (ii) If the queen is drawn and put aside, what is the probability that the second card picked up is (a) an ace? (b) a queen? **[3]**

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10: Case Study Analysis

A disaster management cell is coordinating search and rescue operations for a missing helicopter that is reported to have crashed somewhere in a rectangular region of length 9 km and breadth 4.5 km. Inside this rectangular territory, there is a freshwater lake of length 3 km and width 2.5 km. The rest of the area is dense forest terrain.

Based on the above scenario, answer the following:

(i) Calculate the total geographical area of the search region. **[1]**

(ii) Calculate the surface area of the lake. **[1]**

(iii) What is the probability that the helicopter crashed inside the lake? **[1]**

(iv) What is the probability that the helicopter crashed in the forest region outside the lake? **[1]**

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